Empirical Geometry of Organizational Systems: A Gradient-Based Analysis of 28 Heterogeneous Complex Systems

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1 Abstract

A companion paper introduced a gradient-based classification framework for heterogeneous complex systems, defining five organizational axes and seven morphodynamic regimes. The present work tests this framework empirically on a corpus of 28 systems spanning seven domains: biological, technological, institutional, economic, social, scientific, and infrastructure. Systems range from a NaCl crystal to the European Central Bank, from Occupy Wall Street to a eukaryotic cell.

Four hypotheses are evaluated. First, that the organizational space has low intrinsic dimensionality: under the current scoring, principal component analysis suggests this, with three components explaining 90% of the variance (participation ratio = 3.05). Second, that inter-axis gradients are more structurant than absolute axis values: the gradient Δ_{23} (propagation minus integration) structures the second principal component, and gradient ranks are perfectly stable under scoring perturbation ($\sigma = 0.05$). Third, that natural clusters correspond to the classifier's regimes: four morphodynamic families emerge from unsupervised clustering (adjusted Rand index = 0.648 against regime labels), interpretable as *fluids*, *propagatives*, *organized*, and *rigids*. The classifier defines constraint-based regime boundaries, whereas clustering identifies similarity-based groupings; their partial divergence is therefore expected and informative. Fourth, that the equilibrium–individuation frontier (F01/F06) is a fundamental transition: this is partially confirmed, but two orthogonal transition axes are identified rather than one.

The analysis also reveals that the correlation between hierarchical depth (A_1) and revision capacity (A_5), initially strong ($r = +0.81$), was an artifact of corpus composition, experimentally demonstrated by adding systems that dissociate these properties. By contrast, the correlation between integration (A_3) and normativity (A_4) persists ($r = +0.74$, Spearman $\rho = 0.65$, $p < 0.001$) and may reflect a structural coupling between coordination and regulation in organizational systems.

The corpus, classifier, scoring notes, and analysis scripts are publicly available.[1]

2 1. Introduction

The companion paper (Author, 2026a) introduced a five-dimensional organizational space in which heterogeneous complex systems are represented by vectors $x = (A_1, \ldots, A_5) \in [0, 1]^5$, and classified into seven morphodynamic regimes defined primarily by gradient relationships between axes. That paper was a method paper: it presented the instrument — axes, gradients, regimes, classifier, diagnostics — and evaluated it on fifteen systems selected for maximal contrast.

[1] `https://github.com/MMli97/1-morphodynamic-index`

The present paper is an empirical paper. Its purpose is to test whether the geometric structure suggested by the framework holds up when the corpus is expanded from 15 to 28 systems, drawn from a wider range of domains, and including systems specifically designed to stress-test the instrument's assumptions.

The central question is:

> Does the organizational property space exhibit a stable low-dimensional geometry, and if so, what structures it?

Four operational hypotheses are derived from this question:

1. **Low intrinsic dimensionality.** The effective dimensionality of the space is 2–3, not 5.
2. **Gradient dominance.** Inter-axis gradients (Δ_{23}, Δ_{45}) discriminate better than absolute axis values.
3. **Natural clusters.** Unsupervised clustering recovers structures that correspond partially to the a priori regime definitions.
4. **Transition structure.** The F01/F06 frontier (equilibrium to individuation) is a principal axis of organizational transition.

These hypotheses are stated with explicit success criteria (Section 3) and evaluated through PCA, clustering, sensitivity analysis, and correlation diagnostics. The corpus design follows a progressive scoring method with intermediate analysis, allowing early detection of artifacts and targeted addition of systems to break suspected correlations.

The remainder of the paper is organized as follows. Section 2 describes the corpus. Section 3 reports the geometric analysis. Section 4 presents the sensitivity and robustness tests. Section 5 discusses the correlation structure and its implications. Section 6 identifies limitations. Section 7 concludes.

3 2. Corpus

3.1 2.1 Design principles

The corpus was constructed in three waves, following a progressive scoring method:

- **Wave 1** (10 systems): re-scored from Paper 1 with a detailed protocol, plus new systems. Purpose: establish calibration baseline.
- **Wave 2** (11 systems): targeted additions to cover low-A_2 and high-Δ_{45} zones. Purpose: break A_2 saturation.
- **Wave 3** (7 systems): targeted additions to break specific correlations (A_1–A_5, A_3–A_4). Purpose: stress-test the correlation structure.

Each system satisfies four scorability criteria: precise delimitation, at least three accessible sources, all five axes evaluable, and no overlap with existing corpus members. Systems with confidence = low on two or more axes were flagged as exploratory and excluded from the primary analysis (none were excluded in practice).

Two scoring corrections were applied after initial scoring:

- **Prussian army** (A_5: $0.80 \to 0.40$): the original scoring captured historical plasticity across 60 years rather than operational plasticity in regime. The corrected score reflects the system's capacity for revision during active functioning, not its biography.

- **PostgreSQL** (A_4: 0.90 → 0.40): the original scoring conflated architectural constraints (ACID, integrity rules) with endogenous normativity. A database does not produce or maintain its own norms — it executes externally imposed constraints. The corrected score applies the Canguilhem distinction (endogenous vs. imposed normativity) consistently.

3.2 2.2 The 28 systems

Table 1: Corpus of 28 systems with corrected scores and classifier assignments.

System	Domain	A1	A2	A3	A4	A5	D23	D45	Regime
Eukaryotic cell	biological	0.90	1.00	1.00	1.00	0.60	0.00	0.40	F01
Global supply chain	economic	0.90	1.00	0.40	0.60	0.70	0.60	-0.10	F03
Static LLM	technological	0.70	1.00	0.60	0.20	0.10	0.40	0.10	F03
Linux kernel	technological	1.00	0.90	0.90	0.90	1.00	0.00	-0.10	F01
Ant colony	biological	0.90	0.90	0.80	0.80	0.50	0.10	0.30	F01
Gut microbiome	biological	0.90	0.90	0.70	0.80	0.80	0.20	0.00	F06
Internet BGP	technological	0.90	1.00	0.60	0.80	0.40	0.40	0.40	F05
European Central Bank	institutional	0.90	1.00	0.80	1.00	0.70	0.20	0.30	F01
Wikipedia	social	0.90	0.90	0.80	0.90	0.80	0.10	0.10	F01
Coral reef	biological	1.00	1.00	0.70	0.80	0.60	0.30	0.20	F01
Temperate forest	biological	1.00	1.00	0.80	0.80	0.80	0.20	0.00	F06
Local market	economic	0.90	0.80	0.70	0.80	0.40	0.10	0.40	F05
Bitcoin blockchain	technological	0.80	0.90	0.90	0.80	0.40	0.00	0.40	F05
Bio neural network	biological	1.00	1.00	1.00	1.00	0.90	0.00	0.10	F01
Benedictine order	institutional	0.90	0.90	0.90	1.00	0.70	0.00	0.30	F01
Electrical grid	infrastructure	1.00	1.00	0.90	0.90	0.60	0.10	0.30	F01
Industrial pipeline	economic	0.80	0.80	0.80	0.90	0.40	0.00	0.50	F05
US Constitution	institutional	0.90	1.00	0.80	1.00	0.70	0.20	0.30	F01
Bureaucracy	institutional	0.90	0.70	0.80	1.00	0.50	-0.10	0.50	F05
Institutional archive	institutional	0.80	0.70	0.80	0.90	0.50	-0.10	0.40	F05
Domestic thermostat	technological	0.70	0.70	0.90	0.70	0.10	-0.20	0.60	F05
NaCl crystal	scientific	0.50	0.80	0.60	0.50	0.20	0.20	0.30	F01
Occupy Wall Street	social	0.60	0.90	0.40	0.30	0.70	0.50	-0.40	F03
PostgreSQL (corr.)	technological	0.90	0.90	0.90	0.40	0.70	0.00	-0.30	F01
Startup in pivot	economic	0.60	0.80	0.50	0.50	1.00	0.30	-0.50	F03
Prussian army (corr.)	institutional	0.90	0.90	0.80	1.00	0.40	0.10	0.60	F05
Parliament in crisis	institutional	0.90	1.00	0.30	0.20	0.20	0.70	0.00	F03
Artist collective	social	0.60	0.80	0.40	0.40	0.80	0.40	-0.40	F03

Regime distribution: F01 = 12, F03 = 6, F05 = 8, F06 = 2. Inside ($d = 0$): 13/28. Near-inside ($d \leq 0.10$): 11/28. Outside: 4/28.

3.3 2.3 Domain coverage

The corpus covers seven domains: biological (7 systems), technological (7), institutional (7), economic (4), social (3), scientific (1), and infrastructure (1). The imbalance toward the first three

domains is deliberate: biological, technological, and institutional systems provide the widest morphodynamic contrasts.

4 3. Geometric Analysis

4.1 3.1 Principal component analysis

PCA was performed on the standardized 28×5 matrix of axis scores.

Table 2: PCA on axis scores (A1–A5). Three components explain 90\% of variance.

Component	Variance	Cumulative	Dominant loadings
PC1	48.1%	48.1%	A4 (+0.55), A1 (+0.55), A3 (+0.51)
PC2	25.0%	73.1%	A2 (+0.75), A3 (-0.42), A4 (-0.33)
PC3	16.9%	90.0%	A5 (+0.90)
PC4	5.9%	95.9%	
PC5	4.1%	100%	

The participation ratio is 3.05, indicating an effective dimensionality near 3. The 70% threshold is reached at PC2. These results should be interpreted cautiously given the small sample size ($N = 28$) and the constructed nature of the axes.

The three components admit interpretations:

- **PC1 — Normed organizational complexity.** High A_4, A_1, A_3: systems that are deep, integrated, and normative. This separates the NaCl crystal and the static LLM (left) from the biological neural network and Linux kernel (right).
- **PC2 — Propagation vs. integration/normativity.** High A_2, low A_3 and A_4: this is essentially the gradient Δ_{23} extended with a normative deficit. The parliament in crisis and supply chain (top) oppose the thermostat and bureaucracy (bottom).
- **PC3 — Revision capacity (pure).** The loading of A_5 on PC3 is +0.898 — nearly a single-axis component. This separates the startup ($A_5 = 1.00$) and Linux ($A_5 = 1.00$) from the thermostat ($A_5 = 0.10$) and the LLM ($A_5 = 0.10$).

A parallel PCA on the three primary gradients (Δ_{23}, Δ_{45}, Δ_{12}) yields two components explaining 88.0% of gradient variance: PC1 loads on Δ_{23} (+0.61) and Δ_{12} (-0.58); PC2 loads on Δ_{45} (+0.80). The gradient space is effectively two-dimensional.

4.2 3.2 Distribution in gradient space

The scatter plot reveals a structured distribution rather than a uniform cloud. Four zones are visually distinguishable: a dense central region (F01 systems), an upper-left normative region (F05: thermostat, bureaucracy, Prussian army), a right-side propagative region (F03: supply chain, LLM, parliament), and a lower region with negative Δ_{45} (Occupy, startup, artist collective).

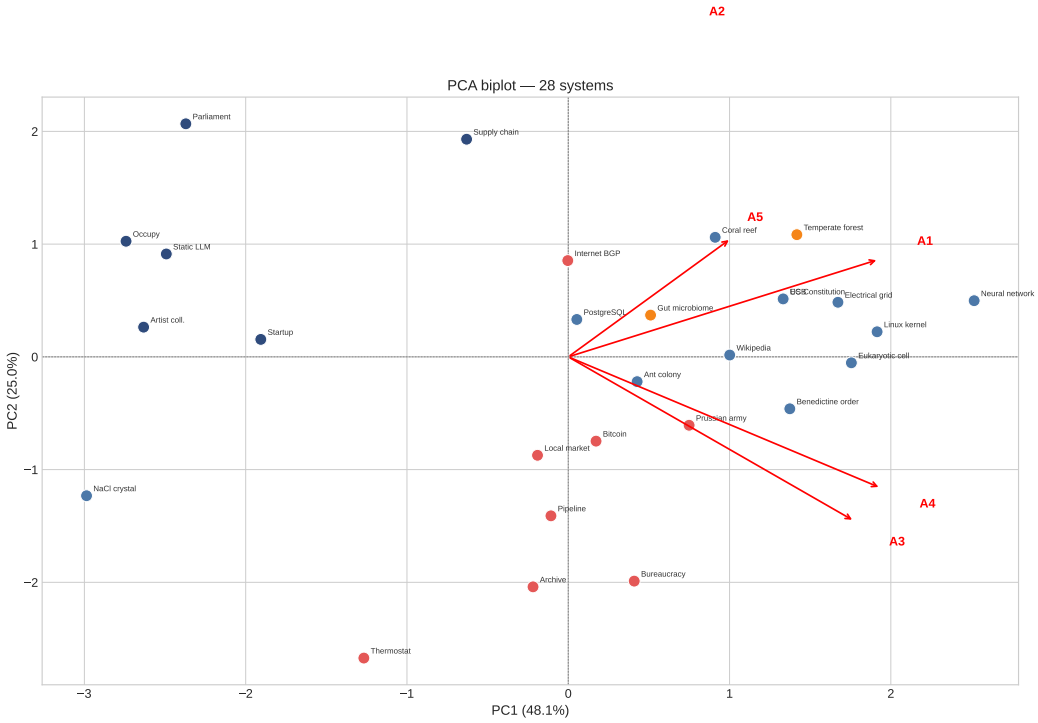

Figure 1: PCA biplot of 28 systems in the PC1–PC2 plane. Loading arrows show axis contributions. Colors indicate regime assignments.

4.3 3.3 Clustering

Unsupervised clustering was performed using both k-means and Ward hierarchical clustering on standardized scores.

Table 3: Clustering results. ARI is computed against classifier regime labels.

Method	k	Silhouette	ARI vs. regimes
k-means	3	0.316	0.544
k-means	4	0.331	0.648
Ward	4	0.319	0.454

At $k = 4$, k-means produces four interpretable families:

- **Organized** (13 systems): cell, Linux, microbiome, ECB, Wikipedia, forest, neural network, Benedictine order, grid, Constitution, coral reef, PostgreSQL, ant colony. Mean profile: all axes above 0.72.
- **Rigid** (7 systems): local market, Bitcoin, pipeline, bureaucracy, archive, thermostat, Prussian army. Mean profile: $A_5 = 0.39$, $A_4 = 0.87$, $\Delta_{45} = +0.48$.
- **Propagative** (4 systems): supply chain, LLM, BGP, parliament in crisis. Mean profile: $A_2 = 1.00$, $A_3 = 0.48$, $\Delta_{23} = +0.52$.

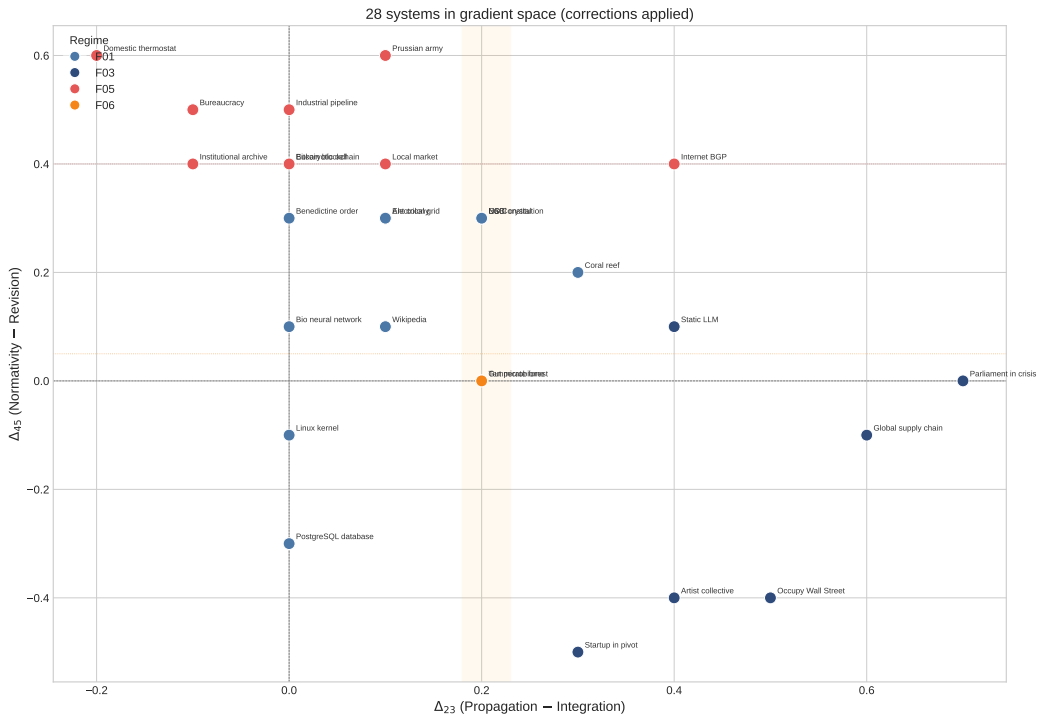

Figure 2: Distribution of 28 systems in the gradient plane $(\Delta_{23}, \Delta_{45})$. Colors indicate classifier regime.

- **Fluid** (4 systems): NaCl crystal, Occupy, startup, artist collective. Mean profile: $A_1 = 0.58$, $A_5 = 0.68$, $\Delta_{45} = -0.25$.

The correspondence between clusters and classifier regimes is partial but substantial: propagatives correspond to F03, rigids to F05, fluids to a mix of F03 and F06. The organized cluster is dominated by F01. The ARI of 0.648 exceeds the pre-registered threshold of 0.3 but falls short of identity. This mismatch is informative: clustering recovers global profile similarity, while the classifier captures gradient-based transitions not directly recoverable from profile similarity alone. The two methods measure overlapping but distinct aspects of the organizational space.

5 4. Robustness

5.1 4.1 Sensitivity to scoring noise

Each axis score was perturbed by Gaussian noise (N draws $= 500$) at two levels: $\sigma = 0.05$ (half a sub-criterion) and $\sigma = 0.10$ (one full sub-criterion).

Table 4: Sensitivity results. Baseline: k-means k = 4.

Metric	sigma = 0.05	sigma = 0.10
ARI vs. baseline clusters	0.679	0.490
P(ARI > 0.5)	92.2%	46.0%
D23 rank stability (rho > 0.8)	100%	66.6%

Metric	sigma = 0.05	sigma = 0.10
D45 rank stability (rho > 0.8)	100%	98.6%

At $\sigma = 0.05$, the structure is robust: clusters are preserved in 92% of draws, and gradient ranks are perfectly stable. At $\sigma = 0.10$, cluster stability degrades but gradient ranks partially hold, especially for Δ_{45} (98.6%). Robustness is therefore local ($\sigma = 0.05$, corresponding to half a sub-criterion of scoring uncertainty) but not global ($\sigma = 0.10$). The gradient ordering is more robust than the cluster assignments, suggesting that gradients may capture a more stable aspect of the data structure than absolute profiles.

5.2 4.2 Stability across corpus expansion

The geometric structure was monitored across three corpus sizes:

Table 5: Evolution of geometric indicators across corpus expansion.

Corpus	N	PC1	PC1-2	PC1-3	PR	Silhouette (k=3)
Wave 1	10	60.9%	78.1%	94.0%	2.34	0.247
Waves 1+2	21	51.0%	81.1%	92.3%	2.73	0.339
All (corrected)	28	48.1%	73.1%	90.0%	3.05	0.316

PC1 decreases as the corpus expands (from 61% to 48%), indicating that no single factor dominates. But PC1-3 remains above 90% throughout, and the participation ratio increases steadily, confirming that the space opens toward a balanced three-dimensional structure. This is the expected behavior for a genuine low-dimensional manifold gaining resolution, not for a noise-dominated space.

6 5. Correlation Structure

6.1 5.1 The A_1–A_5 correlation: demonstrated artifact

The Pearson correlation between hierarchical depth (A_1) and revision capacity (A_5) was $r = +0.81$ at $N = 21$, raising concerns about axis redundancy. Wave 3 was designed to test this: systems with high A_1 and low A_5 (NaCl crystal, Prussian army, parliament in crisis) and systems with low A_1 and high A_5 (Occupy, startup, artist collective) were added specifically to probe this correlation.

Result: $r = +0.28$ at $N = 28$ (corrected). The correlation collapsed. Spearman $\rho = +0.34$ ($p = 0.077$, non-significant).

This demonstrates that the initial correlation was a corpus composition artifact: the first 21 systems happened to be drawn from a region of the space where depth and revision covary. The addition of structurally targeted systems broke the correlation without modifying the scoring protocol. Hierarchical depth and revision capacity are empirically independent properties.

6.2 5.2 The A_3–A_4 correlation: possible structural signal

The Pearson correlation between integration (A_3) and normativity (A_4) is $r = +0.74$. Spearman $\rho = +0.65$ ($p < 0.001$). This correlation survived the corpus expansion, the scoring corrections, and the change of correlation method.

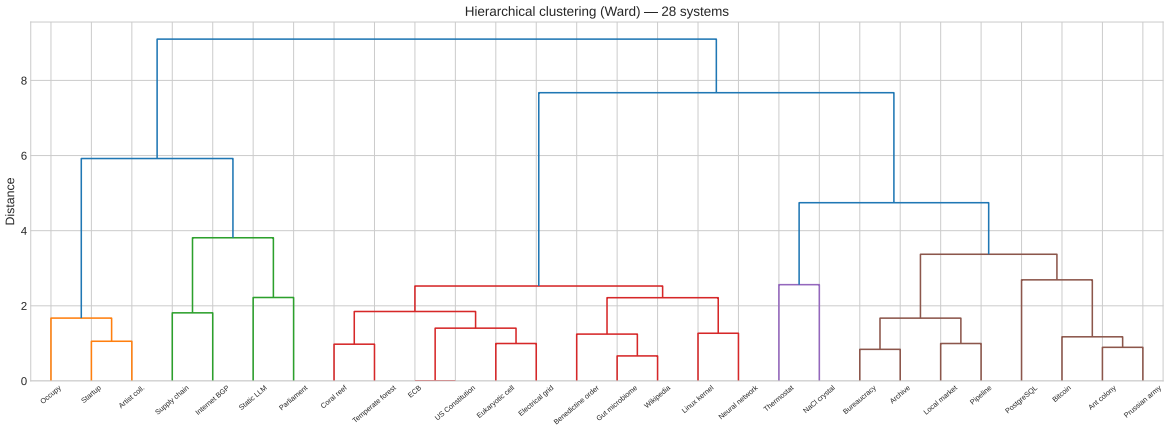

Figure 3: Hierarchical clustering (Ward) dendrogram of 28 systems.

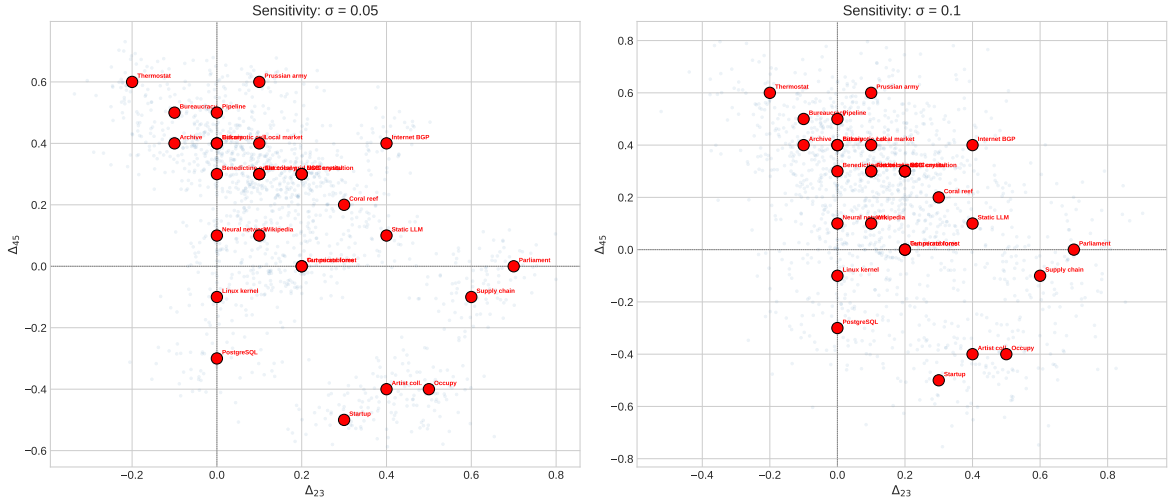

Figure 4: Sensitivity visualization. Red: original positions. Blue clouds: 50 noisy draws. Left: $\sigma = 0.05$. Right: $\sigma = 0.10$.

Two interpretations must be considered:

Structural signal. Systems that coordinate well (A_3 high) tend to be normative (A_4 high) because coordination requires rules, and normativity requires coordination to be enforced. Conversely, systems without coordination (Occupy, parliament in crisis) cannot sustain normativity. If this is correct, the coupling reflects a genuine organizational constraint: integration and normativity are co-produced.

Scoring artifact. Sub-criteria I_5 (maintenance of coherent global state) and N_4 (internal stabilization mechanism) may overlap conceptually: both concern the system's capacity to maintain a stable configuration. If the same empirical evidence is scored under both, the correlation is inflated. The corpus also lacks systems with high A_3 and low A_4: PostgreSQL (corrected to $A_3 = 0.90$, $A_4 = 0.40$) is the only case, and it is insufficient alone to break the correlation. Systems such as drone swarms, slime molds, or peer-to-peer networks without consensus protocols could test this.

We report both interpretations without adjudicating. Resolving this question requires either formal proxy measurements that independently operationalize A_3 and A_4, or the addition of systems

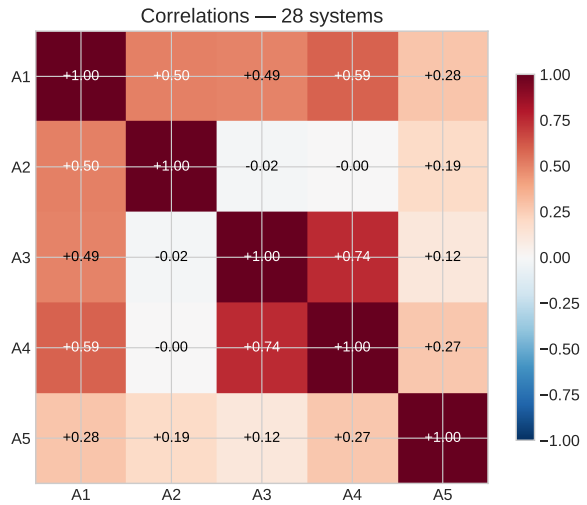

Figure 5: Correlation matrix of axes A1–A5 (Pearson). The A3–A4 coupling is the strongest persistent correlation.

that dissociate these two properties.

6.3 5.3 Variance and saturation

Table 6: Axis statistics across 28 systems.

Axis	Mean	Std	Range
A1	0.85	0.135	0.50–1.00
A2	0.90	0.100	0.70–1.00
A3	0.73	0.187	0.30–1.00
A4	0.75	0.251	0.20–1.00
A5	0.58	0.244	0.10–1.00

A_2 remains the axis with the lowest variance (std = 0.10). This partial saturation is the most important structural limitation of the current dataset: the discriminative power of Δ_{23} currently relies primarily on variation in A_3, since A_2 contributes little discriminating variance. This does not invalidate the gradient, but it means that Δ_{23} measures primarily integration deficit rather than a genuine propagation–integration balance. Resolving this requires the addition of systems with genuinely low propagation capacity.

A_5 has the highest variance (std = 0.244), consistent with its role as the dominant loading on PC3 — the axis of revision captures the widest range of organizational configurations.

7 6. Limitations

1. **Single annotator configuration.** All 28 systems were scored using a single annotator configuration: a human researcher providing sources and interpretive guidance, with an LLM assistant (Claude, Anthropic) applying the scoring protocol in separate sessions. This means the scores reflect the calibration of one specific human–model pair. The scoring corrections

(Prussian army, PostgreSQL) demonstrate internal consistency checking within this configuration, but inter-annotator reliability — whether different humans, different LLMs, or different human–LLM pairs would produce the same geometric structure — remains untested. This is the most serious methodological limitation. Multi-annotator validation using independent LLMs is planned for future work.

2. **Corpus size.** $N = 28$ supports exploratory analysis but not statistical inference. The organized cluster (13 systems) remains too large and heterogeneous. Some regime zones (F08, strict F09) are empty.

3. **Partial A_2 saturation.** The low variance of A_2 (std = 0.10) limits the discriminant power of Δ_{23}. The addition of systems with genuinely low propagation capacity (passive archives, isolated structures) would be needed.

4. **No temporal dynamics.** The corpus consists of instantaneous snapshots. Organizational transitions (rigidification, crisis, individuation) are suggested by the gradient space topology but not directly observed.

5. **Calibration mismatch.** The classifier V2.1.2 was calibrated on the original 15-system battery (lower scores). The expanded corpus (higher scores) shows 4 outside systems. Recalibration (V3) should follow corpus stabilization.

6. **Unresolved A_3–A_4 correlation.** The strongest inter-axis correlation ($r = +0.74$) remains unresolved between structural signal and scoring artifact. The corpus lacks sufficient systems dissociating these properties.

8 7. Conclusion

This work tested the geometric hypotheses generated by an exploratory morphodynamic framework on a corpus of 28 heterogeneous complex systems. Three principal observations emerge.

First, under the current scoring, the organizational space appears to be of low intrinsic dimensionality. Three principal components explain 90% of the variance, and the participation ratio (3.05) suggests an effective dimensionality near three: normed organizational complexity, propagation-integration tension, and revision capacity. This structure is stable across corpus expansion from 10 to 28 systems.

Second, gradient relationships between organizational properties appear more stable and more discriminating than absolute property levels. The gradient Δ_{23} (propagation minus integration) is the strongest structural discriminant, with rank stability of 100% under scoring perturbation at $\sigma = 0.05$. This suggests that relational structure (gradients) is more stable than absolute positioning in organizational property space. However, the discriminative power of Δ_{23} currently relies primarily on variation in A_3 due to partial saturation of A_2. Four morphodynamic families — fluids, propagatives, organized, and rigids — emerge from clustering and correspond partially to the classifier's a priori regime definitions, though clustering captures global profile similarity rather than the gradient-based transitions that define regimes.

Third, the correlation structure of the axes is partly artifactual and partly structural. The A_1–A_5 correlation ($r = +0.81$) was broken experimentally by adding systems that dissociate depth and revision, demonstrating that corpus design can distinguish genuine inter-property couplings from sampling biases. The A_3–A_4 correlation ($r = +0.74$) resists all attempts to weaken it and may reflect a real organizational constraint: that coordination and regulation are co-produced. This

remains an open question. More broadly, the observed geometric structures persist despite targeted attempts to break them through corpus design — a form of robustness that goes beyond statistical stability under noise.

These results should be understood as properties of the current instrument and dataset, not as established properties of complex systems in general. The framework is an exploratory instrument whose adequacy is evaluated through diagnostic tests, not a validated theory.

Three directions for future work follow directly. First, multi-annotator validation using independent LLM scorers to test whether the geometric structure depends on the annotator configuration. Second, the addition of systems dissociating integration and normativity to resolve the A_3–A_4 question. Third, the exploration of organizational transitions through temporal scoring of systems at different life-cycle stages.

The complete dataset, classifier code, scoring notes, and analysis scripts are available at `https://github.com/MMli97/1-morphodynamic-index`.

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